Let F be a field, and S_n be a Symmetric group on n-letters.
Consider the Group-Action of S_n on $M_{n\times n}(F)$
First, for standard-basis $\{e_1, \dots, e_n\}$ for F^n, we can denote as

$$A = \begin{pmatrix} \underbrace{Ae_1}_{n\times 1} & \cdots & Ae_n \end{pmatrix} = \begin{pmatrix} e_1 A \\ \vdots \\ \underbrace{e_n A}_{1\times n} \end{pmatrix}$$

Now, define the Right Action on $M_{n\times n}$ by S_n as;

$$M_{n\times n} \times S_n \to M_{n\times n} : (A, \sigma) \mapsto \begin{pmatrix} Ae_{\sigma(1)} & \cdots & Ae_{\sigma(n)} \end{pmatrix} \left(\overset{\text{denote}}{=} A_\sigma \right) \left(= A \cdot E_\sigma \right)$$

And similarly, define the Left Action:

$$S_n \times M_{n\times n} \to M_{n\times n} : (\sigma, A) \mapsto \begin{pmatrix} e_{\sigma(1)} A \\ \vdots \\ e_{\sigma(n)} A \end{pmatrix} \left(\overset{\text{denote}}{=} \sigma A \right)$$

And, define the Conjugation Action;

$$S_n \times M_{n\times n} \to M_{n\times n} : (\sigma, A) \mapsto E_\sigma^{-1} \cdot A \cdot E_\sigma . \left(\overset{\text{denote}}{=} \sigma(A) \right)$$

Note: $E_\sigma^{-1} = E_{\sigma^{-1}}$

Prop. For any diagonal Matrix D and permutation $\sigma \in S_n$, $\sigma^{-1} D \sigma$ is diagonal with $\sim D$.

Proof. Let $D = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$ denote the diagonal Matrix.

Let $r = (i\ j) \in S_n$ $(1 \le i < j \le n)$ be a transposition.
Put $E_{i,j} = \begin{pmatrix} e_{r(1)} & \cdots & e_{r(n)} \end{pmatrix}$. Then, by simple calculation, $E_{i,j}^{-1} = E_{i,j}$.

Moreover, for any $A = \begin{pmatrix} \alpha_1 & \cdots & \alpha_n \end{pmatrix}$, $AE_{i,j} = \begin{pmatrix} \alpha_{r(1)} & \cdots & \alpha_{r(n)} \end{pmatrix}$, and similarly

$$E_{i,j} A = \begin{pmatrix} \beta_{r(1)} \\ \vdots \\ \beta_{r(n)} \end{pmatrix} \text{ where } A = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}.$$

Therefore, $$E_{i,j} \cdot D \cdot E_{i,j} = \begin{pmatrix} \lambda_{r(1)} & & \\ & \ddots & \\ & & \lambda_{r(n)} \end{pmatrix} \sim D$$

Finally, since every permutation can be represented by product of transposition, we obtain the result, inductively.